

CHAPTER 1 INTRODUCTION

1.1. BACKGROUND OF RESEARCH

Bike-sharing systems (BSSs) have been deployed in many cities worldwide as a flexible form of urban mobility and as part of environmentally sustainable transport systems (Alvarez-Valdes et al., 2016). As of the end of 2024, there were 9,278,605 shared bicycles operating in 2,145 systems across over 1,700 cities in 92 countries (The Meddin Bike-sharing World Map, 2024). As a substitute for motorized travel, bike sharing helps relieve traffic congestion and reduce greenhouse gas and pollutant emissions (Kou et al., 2020; Luo et al., 2019), and it is well suited to short trips and to connections with public transit, including first- and last-mile travel (DeMaio, 2009; Jin et al., 2025; Nath & Rambha, 2019).

Based on how shared bicycles are locked and parked, BSSs are commonly classified into station-based systems, where bicycles are parked at designated docking stations, and free-floating systems, where bicycles are equipped with automatic locks and can be parked relatively freely. This thesis focuses on station-based systems. In a station-based BSS, a user rents a bicycle at one station and returns it at another. Consequently, a bike-sharing trip can fail in two ways: the user finds no bicycle to rent at the origin, or no vacant locker in which to return one at the destination (Alvarez-Valdes et al., 2016; Raviv et al., 2013).

These failures arise because rental and return demand vary across stations and over time (Datner et al., 2019). The resulting spatio-temporal imbalance drives some stations toward a shortage of bicycles and others toward a shortage of vacant lockers, leaving arriving users unable to complete their trips (Brinkmann et al., 2016). To reduce the service failures, operators redistribute bicycles among stations using a fleet of dedicated vehicles, an operation known as bike repositioning (Raviv et al., 2013). Generally, bike repositioning can be divided into two types: static and dynamic repositioning. Static repositioning is carried out during an off-peak period,

typically overnight when the system is nearly idle and demand during the operation is negligible (Raviv et al., 2013). Dynamic repositioning is instead performed during the day to react to evolving demand (Caggiani et al., 2018). The problem of determining the route of each vehicle together with the quantities of bicycles loaded or unloaded at the visited stations is known as the bike repositioning problem (BRP) (Brinkmann et al., 2016; Raviv et al., 2013). In particular, this thesis studies the static BRP.

In addition to the usable bicycles handled in conventional repositioning, operators also need to deal with broken bicycles, which cannot be rented to satisfy rental demand. Meanwhile, a broken bicycle parked at a station occupies a locker that could otherwise accommodate a returned bicycle. Broken bicycles are a recurring concern across BSSs worldwide. In Barcelona, about 12% of the bicycles available at stations were reported to have major defects that prevented normal use, and a further 55% had minor defects such as broken bells and lights (Castro Fernández, 2011). Similar patterns are observed elsewhere: an inspection of BIXI Montreal found roughly one in five bicycles in a state of disrepair (DeMaio & Meddin, 2009), while in New York's Citi Bike system about 2% of the fleet requires major repairs every day (Kaspi et al., 2017).

To handle broken bicycles, operators commonly collect them and transport them to a depot for repair, thereby freeing occupied lockers at stations. Some broken bicycles may also be repaired on site as part of routine maintenance, restoring them to service without depot transport. For example, Vélib' in Paris employs maintenance agents who repair bicycles on the street rather than transporting them to repair shops (Tironi, 2011), and Bicing in Barcelona deploys rapid-intervention motorbikes for minor repairs at stations (Clear Channel International, 2010). In practice, therefore, operators need to coordinate the repositioning of usable bicycles with the collection of broken bicycles and, where possible, their repair on site.

These operations jointly determine the resulting numbers of usable bicycles

and vacant lockers at stations for the next day, and hence the system’s ability to satisfy subsequent rental and return demand. In real systems, such decisions are made for large networks. In April 2026, for example, New York City’s Citi Bike operated about 2,352 active docking stations and a fleet of more than 35,000 bicycles (Citi Bike, 2026). Coordinating the repositioning of usable bicycles with the collection and on-site repair of broken ones, at this scale and under demand that varies across stations and over time, poses a substantial operational challenge for system operators.

1.2. MOTIVATIONS AND OBJECTIVES

Quantifying how well a station will serve its users on the next day requires a measure of the dissatisfaction that arises when its bicycles or lockers run short. In this thesis, such evaluation is operationalized through two service-level measures. The user dissatisfaction function (UDF) (Raviv & Kolka, 2013) expresses the expected shortages of bicycles and lockers at a station as a function of its initial inventory, while the extended user dissatisfaction function (EUDF) (Kaspi et al., 2017) further accounts for the initial inventory of broken bicycles. Based on these service-level measures, this thesis is motivated by three research gaps in static bike repositioning.

First, UDF-based static bike repositioning remains difficult to solve at large station-network scales. The problem jointly determines station-level bicycle movements and vehicle routes, and its search space grows rapidly with the numbers of stations and vehicles. Clustering-based decomposition can ease this burden by reducing the network size before detailed routing is performed (Forma et al., 2015). Existing decompositions form clusters mainly from geographic and inventory considerations (Forma et al., 2015) or from service-level feasibility and approximate routing costs (Schuijbroek et al., 2017). The estimated UDF reduction attainable within a cluster and the per-vehicle routing-time feasibility have not yet been considered together at the clustering stage, so the resulting partition may be misaligned

with both service improvement and route feasibility.

Second, broken bicycles create additional operational work that needs to be planned jointly with usable-bike repositioning. Some broken bicycles may be collected by trucks and transported to a depot, while others may be repaired on site by repairers. This workload is substantial. Citi Bike’s publicly available monthly operating reports record the number of bicycle repairs completed each month, which, aggregated by calendar year, give the annual repair activity shown in Figure 1-1.

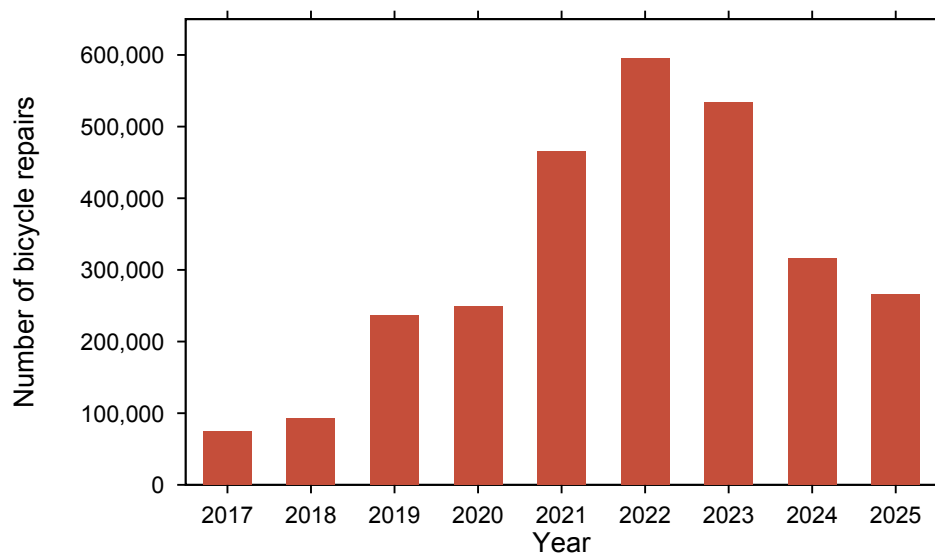


Figure 1-1. Annual bicycle repair activity from 2017–2025. Data obtained from Citi Bike Monthly reports.¹

The reported annual repair activity rose from 74,464 completed bicycle repairs in 2017 to a peak of 594,649 in 2022 and remained substantial at 265,310 in 2025. The workload is also visible in recent monthly operations: Citi Bike reported 11,637 completed depot repairs and 10,565 unique bicycles checked or repaired across depot and field operations in April 2026 alone (Citi Bike, 2026). Handling this workload interacts directly with usable-bike redistribution. Depot collection frees the lockers occupied by broken bicycles, but it also consumes truck capacity that would otherwise serve usable-bike redistribution (Y. Wang & Szeto, 2018). Because the associated truck routes and loads affect both operating effort

¹<https://citibikenyc.com/system-data/operating-reports>

and emissions, broken-bicycle handling also raises an environmental concern. Frequent usable-bike repositioning can substantially increase the fuel consumption of repositioning vehicles, which conflicts with the environmental motivation for operating BSSs as an eco-friendly transport mode (Luo et al., 2019). In London, for example, fossil-fueled vehicle usage for repositioning has been reported to outweigh the environmental benefit of replacing car trips by bike-sharing trips (Fishman et al., 2014). Since repositioning plans with long routes or heavy vehicle loads may increase air-pollutant emissions (Shui & Szeto, 2018), the collection of broken bicycles can further intensify this concern: collected broken bicycles occupy vehicle capacity and have to be carried until the trucks return to the depot. On-site repair introduces an additional intervention, in which repairers restore broken bicycles in the field. The resulting planning problem must coordinate truck-based usable-bike repositioning, truck-based broken-bicycle collection, and repairer-based on-site repair. Previous BRP studies have incorporated broken-bicycle collection into repositioning. However, coordinating usable-bike repositioning, broken-bicycle collection, and repairer-based on-site repair within a single static plan has, to the best of the author's knowledge, not yet been studied.

Third, the integrated problem with broken bicycles and on-site repair requires a scalable decomposition method of its own. The static plan described above is difficult to solve at large station-network scales, so stations must be grouped before detailed truck and repairer routes are optimized. At this clustering stage, however, a candidate cluster can no longer be evaluated only by one-dimensional usable-bike inventory changes. Its value depends on the bivariate state of each station, on the combined effect of usable-bike transfers, broken-bicycle collection, and on-site repair, and on whether the truck and repairer workloads, which proceed in parallel, both fit within a common operating time budget. Existing clustering-based decompositions have not evaluated clusters through bivariate EUDF changes under parallel truck and repairer time demands. This creates a distinct large-scale clustering problem in which service improvement and operational feasibility need to be evaluated over coupled usable-bike and broken-bicycle states.

To fill the abovementioned research gaps, this thesis aims to:

1. To develop a UDF-aware cluster-first, route-second framework for the multi-vehicle static bike repositioning problem, in which estimated UDF reduction and per-vehicle routing-time feasibility are incorporated into the clustering stage, and to evaluate its UDF-based service performance and computational scalability on large station networks.
2. To formulate and solve an integrated static bike repositioning problem that coordinates truck-based usable-bike repositioning and broken-bicycle collection with repairer-based on-site repair, and to examine the operational conditions under which introducing on-site repairers is cost-effective.
3. To develop an EUDF-aware large-scale decomposition framework for static bike repositioning with broken bicycles and on-site repair, in which candidate clusters are evaluated through bivariate EUDF changes and through parallel truck and repairer time components under a common time budget, and to examine how jointly accounting for broken-bicycle handling and repairer operations shapes the resulting clusters and large-scale solution quality.

1.3. SCOPE AND OUTLINE OF THE THESIS

The thesis focuses on station-based bike-sharing systems and static, off-peak planning. It considers multi-vehicle truck operations for usable-bike repositioning and broken-bicycle collection, together with independently routed repairers who perform on-site repairs. The models and solution methods developed in this thesis combine user-dissatisfaction-based service evaluation, mixed-integer linear programming, cluster-first, route-second decomposition, constructive heuristics, and hybrid genetic search.

The remainder of this thesis is organized as follows.

Chapter 2 reviews the literature on static bike repositioning, user-dissatisfaction-based service-level evaluation, broken-bicycle collection, on-site repair operations, and decomposition methods for large-scale bike repositioning problems. It identifies the methodological and operational gaps addressed by the three research objectives.

Chapter 3 develops a UDF-aware cluster-first, route-second framework for the multi-vehicle static bike repositioning problem. It formulates the clustering stage to account for estimated UDF reduction and per-vehicle routing-time feasibility, introduces Activated Transfer Selection (ATS) to evaluate candidate clusters, and develops Hybrid Genetic Search for UDF-aware Clustering (HGS-UC) to solve the large-scale clustering problem.

Chapter 4 studies integrated usable-bike repositioning, broken-bicycle collection, and on-site repair. It presents a time-indexed mixed-integer linear programming formulation that coordinates truck routes, repairer routes, station-level bike-handling quantities, and repair quantities. With truck routes and loads explicitly modeled, the objective also accounts for truck carbon emissions. It develops HGSADC-SBC, which combines Hybrid Genetic Search with Adaptive Diversity Control and a Station-Budget-Constrained heuristic, as the solution method. The numerical analyses examine the operational conditions under which introducing on-site repairers is cost-effective.

Chapter 5 develops an EUDF-aware decomposition framework for large-scale static bike repositioning with broken-bicycle collection and on-site repair. It introduces Extended Activated Transfer Selection (EATS) to estimate cluster-level EUDF reduction while accounting for separate truck and repairer time requirements under a common operating time budget, and Hybrid Genetic Search for Extended-UDF Clustering (HGS-EUC) to solve the resulting large-scale station-partitioning problem. The cluster-level planning instances are then solved by the integrated solver developed in Chapter 4.

Chapter 6 summarizes the principal findings and contributions of the thesis, discusses its limitations, and outlines directions for future research.

CHAPTER 2 LITERATURE REVIEW

2.1. CATEGORIES AND OBJECTIVES IN EXISTING BRP STUDIES

2.1.1. Problem categories and settings

Existing approaches for reducing demand–supply imbalance in bike-sharing systems can be broadly classified into user-based and operator-based repositioning. User-based repositioning encourages users to participate in rebalancing by offering incentives for picking up or returning bicycles at selected locations (Cheng et al., 2021; Pan et al., 2019; Singla et al., 2015; J. Wang & Wang, 2021). This mechanism can help shift part of the repositioning workload to users, but its effectiveness depends on user participation and on the design of the incentive scheme. Operator-based repositioning is therefore adopted more widely in the BRP literature. In this setting, vehicles are dispatched by the system operator to visit stations, pick up bicycles from surplus stations, deliver bicycles to deficit stations, and return to the depot after the repositioning operation (Brinkmann et al., 2016; Raviv et al., 2013).

Operator-based BRPs are commonly divided into static and dynamic variants. In static BRPs, repositioning is performed during an off-peak period, usually when user activities are negligible, and station inventories change mainly through operator interventions (Forma et al., 2015; Raviv et al., 2013; Schuijbroek et al., 2017). In dynamic BRPs, repositioning is performed while the system is operating, so user arrivals and repositioning actions change the station states simultaneously (Brinkmann et al., 2019; Caggiani & Ottomanelli, 2013; Caggiani et al., 2018; Contardo et al., 2012). The dynamic setting is useful for real-time response, whereas the static setting is central to overnight preparation for the following demand period. The studies reviewed in this chapter are therefore centered on static BRPs, with dynamic and free-floating studies discussed when they clarify objective design, broken-bicycle handling, or solution methodology.

BRPs also differ in several operational settings. Station-based systems require bicycles to be parked in lockers at designated stations, while free-floating systems allow bicycles to be parked more flexibly within a service area. A repositioning model may use a single vehicle or multiple vehicles, a single depot or multiple depots, homogeneous or heterogeneous fleets, and single-visit or multiple-visit routing assumptions (Alvarez-Valdes et al., 2016; Du et al., 2020; Y. Liu et al., 2018). In most vehicle-based formulations, the key decisions include the route of each vehicle and the number of bicycles loaded or unloaded at each visited station. These routing and quantity decisions are strongly coupled: a route determines the sequence in which inventory can be moved, while the quantity decisions determine vehicle loads, station inventories, handling times, and the final service level.

2.1.2. Objectives in BRP formulations

The objectives of existing BRP studies can be grouped according to operator-oriented and user-oriented concerns. Operator-oriented objectives typically minimize the operational effort of repositioning, such as total travel distance, travel time, routing cost, or vehicle operating cost (Benchimol et al., 2011; Chemla et al., 2013; Dell'Amico et al., 2016). Some studies also consider loading and unloading time or handling cost because these activities can account for a substantial part of the repositioning duration, especially when many bicycles are moved (Rainer-Harbach et al., 2015; Raviv et al., 2013; Szeto et al., 2016). Other operator-side measures include makespan, route balance, fleet utilization, fuel consumption, and emissions (Alvarez-Valdes et al., 2016; Chen & Szeto, 2023; Schuijbroek et al., 2017; Shui & Szeto, 2018; Y. Wang & Szeto, 2018).

User-oriented objectives measure the service consequences of station imbalance. A common approach is to require each station to reach a target inventory level or fall within an acceptable inventory interval after repositioning. When such requirements are relaxed, service level is often represented in the objective by absolute deviations from target inventories, squared deviations, total unmet demand,

satisfied demand, or station-level penalty costs (Brinkmann et al., 2016; Contardo et al., 2012; Di Gaspero et al., 2013; Ho & Szeto, 2014; Rainer-Harbach et al., 2015). These measures are useful because they produce simple station-level quantities that can be combined with routing decisions. They also allow the operator to trade off service level and repositioning cost through weighted-sum objectives (Chen & Szeto, 2023; Forma et al., 2015; Ho & Szeto, 2017; Raviv et al., 2013).

The limitation of target-based and deviation-based measures is that they do not directly evaluate the expected number of users who fail to rent a bicycle or fail to return one. The same absolute deviation from a target inventory can have different service effects at stations with different capacities, rental-return patterns, and demand intensities. A shortage at a high-demand station during a peak period can be more harmful than the same shortage at a low-demand station. This limitation led to a stream of studies that evaluate service quality through user dissatisfaction rather than through a uniform inventory deviation.

2.1.3. User-dissatisfaction-based service evaluation

Raviv and Kolka (2013) introduced the User Dissatisfaction Function (UDF) to evaluate the service quality of a bike-sharing station. A station is modeled as a finite-capacity queueing system with time-varying rental and return arrivals. Given a starting inventory, the UDF gives the expected number of users who fail to rent a bicycle or fail to return a bicycle over a planning horizon. The function is station-specific because it depends on station capacity and on the rental and return arrival processes at that station. It is also inventory-dependent, so changing the post-repositioning inventory of a station changes its expected service failures.

The UDF is well suited to static repositioning because the final inventory after overnight repositioning becomes the initial inventory for the following demand period. In a UDF-based static BRP, the operator does not only decide whether a station is close to a target level; it evaluates how a change in inventory affects expected

rental and return failures. Raviv and Kolka (2013) also showed that the UDF is convex over feasible inventory levels, which provides useful structure for evaluating marginal changes in station inventories.

Several BRP studies have incorporated UDF-based service evaluation. Raviv et al. (2013) formulated static repositioning models that minimize a weighted combination of total UDF and vehicle travel time. Forma et al. (2015) used the UDF in a three-step matheuristic in which clustering, routing, and restricted re-optimization are combined. Ho and Szeto (2017) adopted UDF-based penalties in a hybrid large-neighborhood-search framework. In a dynamic setting, D. Zhang et al. (2017) incorporated UDF into a time-space network flow model and a rolling-horizon solution framework. These studies show that UDF provides a direct service-oriented basis for comparing repositioning plans, but they also show that UDF-based BRPs are computationally demanding because the service objective depends on final station inventories that are jointly determined by routing, loading, unloading, time-budget, and vehicle-capacity constraints.

2.2. BRPS WITH BROKEN BICYCLES AND REPAIR OPERATIONS

2.2.1. Broken-bicycle collection and off-site repair

Broken bicycles change the conventional BRP because they cannot satisfy rental demand and, in station-based systems, may occupy lockers needed for returns. The presence of broken bicycles therefore affects both usable-bicycle availability and locker availability. Early work on this topic focused on identifying unusable bicycles and evaluating their effect on service quality (Kaspi et al., 2016, 2017). Subsequent optimization studies incorporated broken bicycles into routing and repositioning decisions.

One group of studies focuses mainly on broken-bicycle collection or recycling. In such problems, the main task is to route vehicles to locations containing bicycles

that require repair and to transport them to a depot, repair center, or bike shop (H. Lu et al., 2019; S. Zhang et al., 2018). Another group jointly considers usable-bicycle repositioning and broken-bicycle collection, so that the same truck fleet may redistribute usable bicycles and collect broken bicycles within the same planning horizon (Alvarez-Valdes et al., 2016; Cai et al., 2022; Du et al., 2020; Y. Wang & Szeto, 2018, 2021; D. Zhang et al., 2020). These models extend conventional BRPs by introducing a second bicycle type, additional loading or collection quantities, and the need to allocate truck capacity between usable bicycles and broken bicycles.

The common operating assumption in this stream is that broken bicycles are repaired off site. Under this assumption, a broken bicycle must first be collected by a truck before it can be restored to service. Truck collection frees the locker occupied by the broken bicycle, but it does not increase the number of usable bicycles at the station where the broken bicycle was collected. The final station state is therefore determined by how many usable bicycles are delivered or picked up and how many broken bicycles are removed. This off-site repair logic captures an important maintenance operation, but it does not represent field repair as an operational decision.

Broken-bicycle handling has also been studied in dockless and free-floating systems, where usable and faulty bicycles may be scattered over a service area rather than parked at fixed stations. These studies have considered relocation, collection, and recycling decisions under dispersed bicycle locations (S. Chang et al., 2018; X. Chang et al., 2021, 2023; Usama et al., 2019, 2020). Although the physical setting differs from station-based systems, the same modeling issue remains: damaged bicycles are not only maintenance objects; they also affect the logistics of service recovery.

2.2.2. Objectives and service evaluation with broken bicycles

BRPs with broken bicycles have used several types of objectives. Some studies minimize travel cost, travel time, or routing cost while requiring usable bicycles to be balanced and broken bicycles to be collected (Alvarez-Valdes et al., 2016; H. Lu et al., 2019; D. Zhang et al., 2020; S. Zhang et al., 2018). Other studies relax perfect balance and introduce service-related objectives, such as deviations from target usable-bike inventories, satisfied demand, or penalties for unhandled broken bicycles (Cai et al., 2022; Y. Wang & Szeto, 2021; Xu et al., 2019). Green BRP studies further include fuel consumption or carbon emissions, which become relevant when trucks carry usable and broken bicycles over long distances (X. Chang et al., 2023; Y. Wang & Szeto, 2018, 2021).

These objectives capture important aspects of cost and inventory balance, but many of them treat unmet usable-bike demand and unhandled broken bicycles through uniform deviations or penalties. This treatment does not fully reflect the station-specific effect of broken bicycles on user dissatisfaction. A broken bicycle at a low-demand station and a broken bicycle at a high-demand station may have different consequences. Similarly, removing a broken bicycle from a nearly full station can reduce return failures by freeing a locker, whereas removing one from a station with many vacant lockers may have a smaller immediate service effect.

The Extended User Dissatisfaction Function (EUDF) addresses this issue by extending the UDF to a bivariate station state. Kaspi et al. (2017) modeled user dissatisfaction as a function of both usable-bicycle and broken-bicycle inventories. The usable-bicycle inventory affects rental availability, while the broken-bicycle inventory affects effective locker availability and thus return failures. The EUDF is therefore a natural service measure for station-based BRPs with broken bicycles. It links service evaluation to the joint final state of usable and broken bicycles rather than to one inventory type or to a uniform penalty for unhandled broken bicycles.

2.2.3. On-site repair and integrated operations

Off-site repair is not the only way to handle broken bicycles. Some repair work can be performed in the field by repairers who visit stations directly. On-site repair changes the operational role of broken bicycles: a repairer can convert a broken bicycle into a usable bicycle at its current station without using truck capacity to transport it to a depot. This action has a different inventory effect from broken-bicycle collection. Repair reduces the broken-bicycle inventory and increases the usable-bicycle inventory at the same station, whereas truck collection reduces the broken-bicycle inventory but does not create a usable bicycle at that station.

Maintenance and repair decisions have been studied in related sharing-economy and bicycle-sharing contexts. Freund (2018) studied transportation models in the sharing economy, including repair-related operations, and L. Lu et al. (2022) considered predictive-maintenance task assignment in free-floating bicycle-sharing systems. These studies show that maintenance resources need to be assigned and scheduled, but they do not fully integrate repairer routing with truck-based usable-bicycle repositioning and broken-bicycle collection in a single static station-based BRP.

An integrated static BRP with broken bicycles and on-site repair must coordinate two types of mobile resources. Trucks move usable bicycles and collect broken bicycles, subject to vehicle capacity, travel time, loading time, and unloading time. Repairers travel among stations and spend repair time converting broken bicycles into usable bicycles. The two resource types operate in parallel within the same repositioning horizon, and their actions interact through the final usable-bicycle and broken-bicycle inventories at each station. This integrated setting requires the model to determine truck routes, repairer routes, usable-bicycle loading and unloading quantities, broken-bicycle collection quantities, and on-site repair quantities together.

2.3. SOLUTION METHODS AND DECOMPOSITION FOR BRPS

2.3.1. Exact methods and mathematical programming approaches

BRPs are commonly formulated as mixed-integer optimization problems that combine routing decisions with station-level loading and unloading decisions. A solution determines which stations are visited by each vehicle, the visiting order, the quantity loaded or unloaded at each station, and the resulting vehicle loads and final station inventories. Exact methods and mathematical programming approaches are useful because they provide benchmark solutions, clarify the structure of the problem, and can solve small instances under selected formulations.

Several exact or decomposition-based optimization approaches have been used in the BRP literature, including branch-and-bound, branch-and-cut, column generation, Benders decomposition, time-space network formulations, and mixed-integer programming solved by commercial solvers (Brinkmann et al., 2016; Caggiani & Ottomanelli, 2012; Contardo et al., 2012; Dell'Amico et al., 2016; Raviv et al., 2013). These methods are valuable for small networks and for validating heuristic solutions. Their direct use, however, becomes difficult as the number of stations, vehicles, and time or route decisions increases. The computational difficulty is intensified when loading and unloading quantities are decision variables rather than fixed parameters, because the routing decision and the inventory transition along the route have to be optimized together.

The difficulty is greater in UDF- and EUDF-based models. When the objective is a simple travel cost or target deviation, the contribution of a station can often be evaluated by a relatively simple inventory difference. When the objective is UDF, the value of a plan depends on the nonlinear expected service failures produced by the final usable-bicycle inventory at each station. When the objective is EUDF, the final state is bivariate and depends on both usable-bicycle and broken-bicycle inventories. This increases the cost of evaluating a route and makes large-scale exact

optimization particularly challenging.

2.3.2. Heuristics, metaheuristics, and hybrid methods

Heuristics and metaheuristics have been widely developed to obtain high-quality BRP solutions within practical computing times. Existing methods include constructive heuristics, tabu search, variable neighborhood search, large neighborhood search, destroy-and-repair methods, chemical reaction optimization, artificial bee colony algorithms, memetic algorithms, genetic algorithms, and hybrid exact–heuristic frameworks (Ho & Szeto, 2017, 2014; Y. Liu et al., 2018; Y. Lu et al., 2020; Rainer-Harbach et al., 2015; Szeto et al., 2016; Y. Wang & Szeto, 2021). Some methods solve routing and loading decisions together, while others decompose the problem into a routing subproblem and a quantity-assignment subproblem.

Separating routing and quantity decisions can improve computational efficiency. Given a route, the loading and unloading quantities may be determined by linear programming, network-flow methods, dynamic programming, greedy rules, or tailor-made loading heuristics (Forma et al., 2015; Li et al., 2016; Rainer-Harbach et al., 2015; D. Zhang et al., 2017). This route-first, quantity-second logic is useful because candidate routes can be searched by a metaheuristic while station-level quantities are evaluated by a more specialized procedure. The quality of the overall method, however, depends on whether the quantity evaluator is aligned with the objective. A loading rule designed for target deviations may not capture UDF-based service improvement, and a rule designed for usable bicycles alone may not capture the joint EUDF effect of usable-bicycle movement, broken-bicycle collection, and repair.

Hybrid methods combine the strengths of exact and inexact procedures. For example, a heuristic may construct a route and then use an exact or approximate subroutine to determine loading quantities; a metaheuristic may explore routing structures while a problem-specific evaluator estimates the service improvement of

a candidate solution. This pattern is especially relevant to large UDF- and EUDF-based BRPs, where exact optimization of the full model is usually impractical but exact or structured evaluation can still be useful within a heuristic framework.

2.3.3. Cluster-first, route-second decomposition

Cluster-first, route-second (CFRS) decomposition is a common strategy for improving scalability. The station network is first partitioned into smaller groups, and routing or repositioning subproblems are then solved at the cluster level. This strategy reduces the size of each routing problem and can allow cluster-level problems to be solved independently or in parallel. The main challenge is that clustering is not a neutral preprocessing step. A partition that is good for geographic compactness may not be good for inventory transfer, service improvement, or route feasibility.

Existing CFRS and clustering-based studies have used different criteria to form station groups. Schuijbroek et al. (2017) derived inventory bounds from a queueing model and used them in a set-partitioning formulation that accounts for service-level feasibility and approximate routing cost. Forma et al. (2015) proposed a three-step matheuristic in which clustering guides subsequent routing and repositioning decisions. Other studies have used hub or satellite structures, distance and inventory self-sufficiency, fuzzy correlation, feature correlation, or data-driven information to simplify the station network before detailed routing (Huang et al., 2020; J. Liu et al., 2025; Lv et al., 2020, 2022, 2024).

These clustering criteria address different aspects of the BRP. Geographic proximity reduces travel effort. Demand balance and inventory self-sufficiency help construct groups that may internally satisfy repositioning needs. Service-level bounds ensure that station inventories remain acceptable after repositioning. Approximate routing costs help prevent clusters from becoming too dispersed. However, these criteria do not fully evaluate how much user dissatisfaction can be reduced within a cluster under a vehicle time budget. This distinction is important for UDF-based

repositioning because a promising cluster should contain not only nearby stations but also feasible usable-bicycle transfers that generate substantial UDF reduction.

The requirement becomes more demanding in BRPs with broken bicycles and on-site repair. In that setting, a candidate cluster cannot be evaluated by one-dimensional usable-bicycle inventory changes alone. Its value depends on the joint changes in usable-bicycle and broken-bicycle inventories, on the different effects of usable-bicycle pickup and delivery, broken-bicycle collection, and on-site repair, and on whether the truck and repairer workloads can both be completed within the same operating horizon. Existing CFRS methods do not provide this bivariate service-evaluation and two-resource feasibility logic.

2.4. SUMMARY OF RESEARCH GAPS

Table 2-1 summarizes representative BRP studies along the dimensions that are central to this thesis: service evaluation, broken-bicycle handling, repair operations, quantity decisions, and decomposition logic. The rows labelled by thesis chapters are included only to show how the present thesis addresses the gaps; they are not treated as prior literature.

Table 2-1. Summary of representative BRP studies by objectives, operations, quantities, and decomposition features

Reference	Scen.	Sys.	Objectives				Operations				Quantities			Methods/decomposition						
			OC	AD/UD	UDF	EUDF	CE	UB	BB	OR	qUB	qBB	qR	EX	H/M	CF	U-C	E-C	T-C	
Benchimol et al. (2011)	Static	Docked	✓	✓				✓			PB			✓						
Chemla et al. (2013)	Static	Docked	✓					✓			DV			✓	✓					
Raviv et al. (2013)	Static	Docked	✓		✓			✓			DV			✓						
Forma et al. (2015)	Static	Docked	✓		✓			✓			DV				✓	✓	P		P	
Ho and Szeto (2017)	Static	Docked	✓		✓			✓			DV				✓					
Schuijbroek et al. (2017)	Static	Docked	✓	✓				✓			DV			✓	✓	✓			V	
Huang et al. (2020)	Static	Docked	✓	✓				✓			DV				✓	✓				
Lv et al. (2020)	Static	Docked	✓	✓				✓			DV				✓	✓				
Lv et al. (2024)	Static	Docked	✓	✓			✓	✓			DV				✓	✓				
Kaspi et al. (2017)	Static	Docked				✓														
Alvarez-Valdes et al. (2016)	Static	Docked	✓					✓	✓		PB	PB			✓					
Y. Wang and Szeto (2018)	Static	Docked					✓	✓	✓		DV	DV		✓	✓					
S. Zhang et al. (2018)	Static	Docked	✓						✓		NC	PB		✓	✓					
H. Lu et al. (2019)	Static	Dockless	✓						✓		NC	PB			✓					
Usama et al. (2019, 2020)	Static	Dockless	✓	✓				✓	✓		PB	DV		✓	✓					
Du et al. (2020)	Static	Dockless	✓					✓	✓		PB	PB			✓					
D. Zhang et al. (2020)	Static	Docked	✓					✓	✓		PB	PB			✓					
Y. Wang and Szeto (2021)	Static	Docked		✓			✓	✓	✓		DV	DV			✓					
Cai et al. (2022)	Dynamic	Docked	✓	✓				✓	✓		DV	DV		✓	✓					
X. Chang et al. (2023)	Dynamic	Dockless	✓	✓			✓	✓	✓		PB	PB			✓					
Thesis Chapter 3	Static	Docked			✓			✓			DV			✓	✓	✓	✓		V	
Thesis Chapter 4	Static	Docked				✓	✓	✓	✓	✓	DV	DV	DV	✓	✓					
Thesis Chapter 5	Static	Docked				✓		✓	✓	✓	DV	DV	DV		✓	✓		✓	T/R	

Notes: Scen. = scenario; Sys. = system type; OC = operator cost, including travel time, travel distance, routing cost, operating cost, or handling cost; AD/UD = absolute deviation, target-balance, satisfied-demand, or unmet-demand measure; UDF = User Dissatisfaction Function; EUDF = Extended User Dissatisfaction

Function; CE = carbon emissions or fuel consumption; UB = usable-bicycle repositioning; BB = broken-bicycle collection or relocation; OR = on-site repairers; q_{UB} , q_{BB} , and q_R = station-level quantities of usable-bicycle repositioning, broken-bicycle collection, and on-site repair, respectively; DV = decision variable; PB = perfect-balance or complete-collection requirement; NC = not considered; EX = exact or mathematical programming method; H/M = heuristic, metaheuristic, or hybrid method; CF = cluster-first or explicit clustering-based decomposition; U-C = UDF-aware cluster evaluation; E-C = EUDF-aware cluster evaluation; T-C = routing-time feasibility considered in clustering, with V for vehicle-route feasibility and T/R for both truck and repairer feasibility; P indicates partial consideration.

The review identifies three research gaps. First, UDF-based service evaluation and cluster-first decomposition have not been sufficiently integrated in static BRPs. UDF-based models evaluate service quality through expected rental and return failures, while CFRS methods improve scalability by partitioning large station networks before detailed routing. Existing clustering criteria are mainly based on geographic proximity, inventory balance, service-level feasibility, feature correlation, or approximate routing cost. They do not jointly evaluate the UDF reduction attainable within a candidate cluster and the routing-time feasibility of the vehicle assigned to that cluster. Chapter 3 addresses this gap by developing a UDF-aware CFRS framework in which clustering is guided by both estimated UDF reduction and vehicle time feasibility.

Second, existing BRP studies with broken bicycles mainly treat broken bicycles as items to be detected, collected, relocated, or transported for off-site repair. Repair and maintenance tasks are often planned separately from usable-bicycle repositioning. This separation does not capture the static planning setting in which trucks redistribute usable bicycles, collect broken bicycles, and share vehicle capacity between these two tasks, while repairers simultaneously visit stations and repair broken bicycles on site. Existing models also do not jointly determine truck routes, repairer routes, usable-bicycle loading and unloading quantities, broken-bicycle collection quantities, and on-site repair quantities. Chapter 4 addresses this gap by formulating and solving an integrated static BRP that coordinates truck-based usable-bicycle repositioning, truck-based broken-bicycle collection, and repairer-based on-site repair.

Third, the integrated problem with broken bicycles and on-site repair requires a new decomposition logic for large station networks. Cluster quality depends on both usable-bicycle and broken-bicycle inventories, on the combined effects of usable-bicycle movement, broken-bicycle collection, and on-site repair, and on whether truck and repairer workloads can both be completed within the same operating period. Existing CFRS methods do not evaluate clusters through bivariate EUDF

changes while accounting for parallel truck and repairer time requirements. Chapter 5 addresses this gap by developing an EUDF-aware decomposition framework for large-scale static BRPs with broken bicycles and on-site repair.